UNIT - I
Introduction to Automation

Automation and Robotics (23A03606) - handwritten study notes

1. Introduction to Automation

Definition

- **Automation** is the technology by which a process or procedure is performed **without human assistance**, using programmed commands combined with automatic feedback control.
- The word comes from the Greek **automatos** = self-acting. The term was coined by D.S. Harder (Ford Motor Company, 1946).
- A programmed set of instructions + a control system + a power source = an automated system.

> Automation = Power source + Program of instructions + Control system that actuates the instructions.

Need / Reasons for Automation

- Increase **labour productivity** (more output per man-hour)
- Reduce labour cost and shortage of skilled labour
- Mitigate the effects of labour shortage
- Reduce or eliminate routine manual and clerical tasks
- Improve worker **safety** (hazardous, hot, toxic work)
- Improve **product quality** and uniformity
- Reduce manufacturing lead time and work-in-process
- Accomplish processes that cannot be done manually (micro-electronics, high precision)
- Avoid the high cost of not automating (competition)

Advantages and Disadvantages

Advantages	Disadvantages
Higher production rate and productivity	High initial investment
Consistent quality, less scrap	Needs skilled maintenance staff
Lower unit cost in the long run	Unemployment / worker resistance
Safer working conditions	Less flexible for small batches (fixed automation)
Reduced lead time and inventory	Breakdown stops the whole line

Types of Automation

Automated production systems are classified into **three** types:

- **Fixed (hard) automation**
  - Sequence of operations fixed by equipment configuration; changing the product needs new equipment.
  - High initial cost, very high production rate, low unit cost. Suited to **mass production**.
  - Examples: transfer lines, automatic assembly machines, bottling plants.
- **Programmable automation**
  - Equipment can be reprogrammed for a new product; changeover time is spent in reprogramming and setup.
  - Suited to **batch production** with medium volume and variety. Examples: NC machines, industrial robots, PLCs.
- **Flexible (soft) automation**
  - An extension of programmable automation: changeover is almost **zero lost time**, different parts mixed on the same system.
  - Examples: Flexible Manufacturing System (FMS), CNC machines with automatic tool and pallet change.

Feature	Fixed	Programmable	Flexible
Volume	Very high	Medium	Medium
Variety	Very low	Medium-high	High

Investment	High (special)	Medium	Very high
Changeover	Costly	Reprogramming time	Almost none
Production rate	Highest	Medium	Medium

Remember: Fixed ⟶ high volume / low variety. Programmable ⟶ batch. Flexible ⟶ mixed models, no changeover loss.

2. Basic Elements of an Automated System

Every automated system has **three basic elements**:

Power to accomplish the process → Program of instructions → Control system

1. Power to Accomplish the Automated Process

- Electric power is the most common (lighting, motors, controls, computers)
- Also pneumatic and hydraulic power; solar / battery in special cases
- Power is used for the **process** (drive motors, heating) and for **auxiliary functions**: loading/unloading, data processing, communication, material handling.

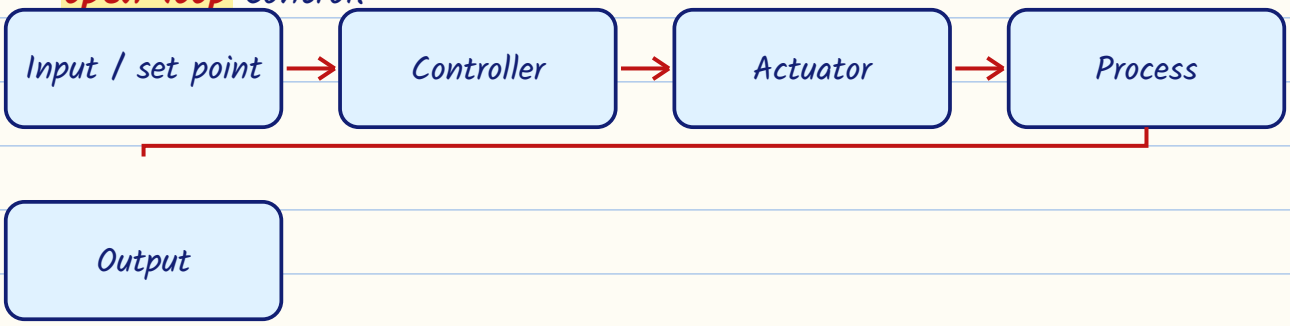
2. Program of Instructions

- The **action program** defines the steps of the process (e.g. a part program for a CNC machine).
- Work cycle program: a sequence of steps for one part (load, clamp, machine, unload).
- Decision-making features: operator interaction, different part styles, variations in raw material.

3. Control System

- Executes the program of instructions: **closed-loop (feedback)** and

open-loop control.



- **Closed loop**: output is measured by a sensor and compared with the reference; the error drives the controller.
- **Open loop**: no feedback; the controller output depends only on the program (e.g. stepper motor drive, washing machine timer).

Open loop	Closed loop
No feedback	Feedback sensor present
Simple and cheap	More complex, costly
Less accurate	Accurate, self-correcting
Stepper motors	Servo motors

3. Advanced Automation Functions

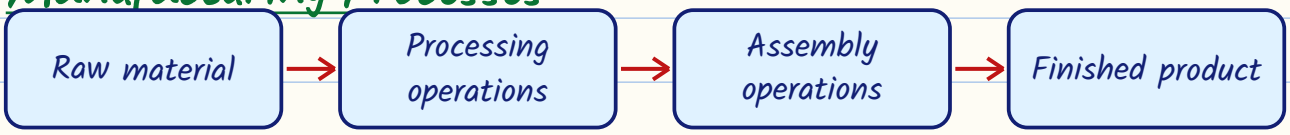
- **Safety monitoring**: detect hazards (fire, collision, overload) and stop or alert.
- **Maintenance and repair diagnostics**: status monitoring, failure diagnosis, preventive maintenance schedule.
- **Error detection and recovery**: error detection, interpretation of the error, and recovery action by the program.

Levels of intelligence: error detection → diagnosis → automatic recovery. Higher the level, fewer human interventions.

4. Manufacturing Industries

- **Manufacturing** = transformation of raw materials into finished products of higher value using physical/chemical processes and machines.
- Industry is classified as:
  - **Primary**: farming, mining, quarrying, oil (use natural resources)
  - **Secondary**: manufacturing of goods, construction, power utilities
  - **Tertiary**: services - banking, retail, education, transport
- Manufacturing industries by product:
  - **Process industries**: chemicals, petroleum, food, paper, cement (continuous or batch processing).
  - **Discrete-item industries**: cars, machines, appliances, electronics (individual parts and assemblies).
  - Products: **consumer goods**, **capital goods** (machine tools, cranes) and **intermediate goods**.

Manufacturing Processes



- **Processing**: shaping (casting, forming, machining), property enhancing (heat treatment), surface processing (cleaning, painting, plating).
- **Assembly**: joining (welding, brazing, soldering), mechanical fastening, adhesive bonding.

5. Types of Production

Production is classified by **quantity** and **product variety**:

Type	Quantity per year	Variety	Layout
Job shop (low)	1 - 100	Very high	Fixed position / process
Batch (medium)	100 - 10,000	Medium	Process or cellular
Mass (high)	10,000 - millions	Low	Product / line layout

- **Job shop production**: one-off or few products, general purpose machines, skilled labour. Eg: prototypes, special machines, ships.
- **Batch production**: lots of the same item produced repeatedly; machines are reset between batches. Eg: books, clothing, machine parts.
  - **Production time of a batch** = setup time + (Q x cycle time).
- **Mass production**: very high volume; two sub types:
  - **Quantity production**: single machine produces one part at a high rate (e.g. stamping).
  - **Flow-line production**: multiple workstations in sequence (e.g. automobile assembly).

Production Quantity vs Variety (P-Q)

- **Product variety**: hard variety (different product types) vs soft variety (small model differences).
- Product variety is inversely related to quantity: more variety ⟶ less quantity per model.

$$\text{Production rate } R_p = 1 / T_c$$

$$\text{Cycle time } T_c = T_o + T_h + T_{th}$$

- T_o = operation time, T_h = handling time, T_{th} = tool handling time.

Production Performance Measures

$$\text{Average production rate } R_p = 60 / T_p$$

$$\text{Availability } A = (MTBF - MTTR) / MTBF$$

$$\text{Utilisation } U = \text{Output} / \text{Capacity}$$

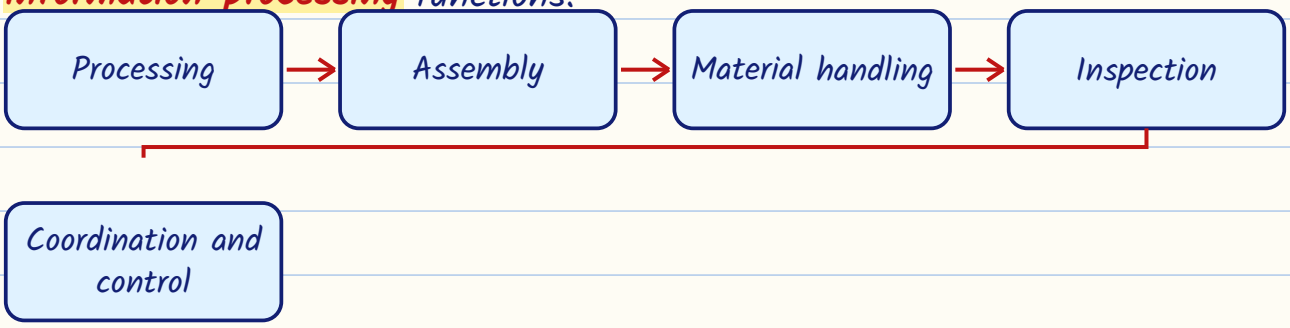
- **Worked example**: T_o = 3 min, T_h = 1 min, T_{th} = 0.5 min ⟶ T_c = 4.5 min ⟶ R_c = 60/4.5 = 13.3 parts/hr.
- If the machine has availability 90%, then actual rate = 13.3 x 0.9 = 12

parts/hr.

MTBF = mean time between failures, MTTR = mean time to repair. Availability falls when repairs take longer.

6. Functions in Manufacturing

Manufacturing has **two groups** of functions: the **physical** and the **information-processing** functions.

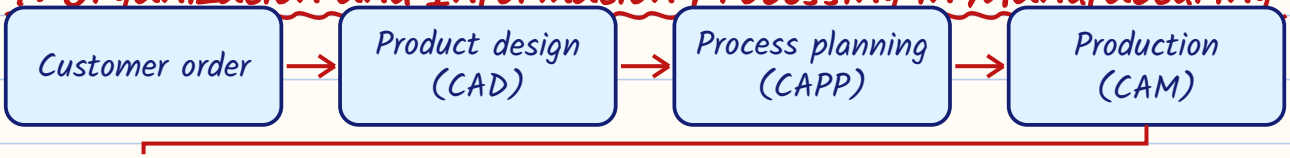


- **Processing and assembly**: the value-adding operations.
- **Material handling and storage**: loading, positioning, transport (AGV, conveyor, forklift), storage (AS/RS).
- **Inspection and test**: quality control, gauging, CMM, vision systems.
- **Coordination and control**: regulation of individual processes (process control) and management of the plant (plant control).

Product Design and Manufacturing Planning

- **Design function**: product concept, CAD, drawings, tolerances.
- **Process planning**: choose operations, machines, tools; route sheet.
- **Production planning**: master schedule, MRP, capacity planning.
- **Quality control**: statistical process control, inspection plans.

7. Organization and Information Processing in Manufacturing



Quality and shipping

- The **information flow** runs in parallel with the physical flow and is managed by computer systems.
- **Product design** ⟶ engineering drawing, bill of materials (BOM).
- **Manufacturing planning**: process planning, master production schedule, MRP, capacity planning.
- **Manufacturing control**: shop-floor control, inventory control, quality control.
- Information processing is organised in **levels**: Enterprise ⟶ Plant ⟶ Cell ⟶ Machine.

CIM: Computer Integrated Manufacturing

- CIM links **CAD, CAM, CAPP, MRP and shop floor control** through a common database and network.
- Benefits: shorter lead time, better quality, lower inventory, faster response.

CAD/CAM = design + manufacturing in one data stream. CIM = CAD/CAM + business functions.

8. Automation Strategies

The **USA principle** (a common sense approach) and the ten strategies for automation:

- **U**nderstand the existing process: input/output, value-added, problems.
- **S**implify the process: eliminate unnecessary steps.
- **A**utomate the simplified process.

The Ten Strategies

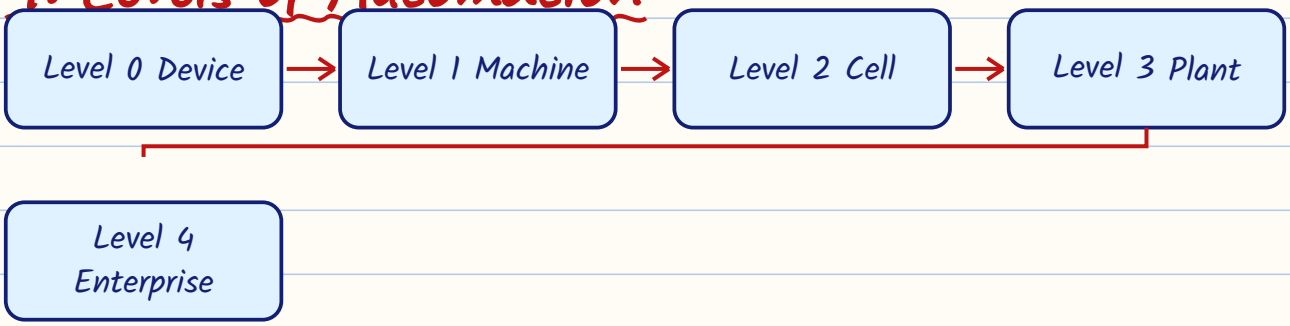
- 1. **Specialization of operations** - dedicated equipment for one task.
- 2. **Combined operations** - several operations on one machine (machining

centre).

- 3. **Simultaneous operations** - perform operations at the same time (multi-spindle drill).
- 4. **Integration of operations** - link workstations with handling, so work moves automatically.
- 5. **Increased flexibility** - quick changeover, programmable machines.
- 6. **Improved material handling and storage** - AGV, AS/RS, conveyors.
- 7. **On-line inspection** - inspect during processing; reduce scrap.
- 8. **Process control and optimisation** - adaptive control, monitor temperature, force etc.
- 9. **Plant operations control** - computer control of inventory, scheduling.
- 10. **Computer integrated manufacturing (CIM)** - integrate all functions.

Mnemonic: S-C-S-I-I-I-O-P-P-C. Learn the keywords of each strategy with one example.

9. Levels of Automation



Level	Name	Examples
0	Device level	Sensors, actuators, motors, valves
1	Machine level	CNC machine, robot, PLC
2	Cell / system level	Manufacturing cell, FMS, assembly line
3	Plant level	Production control, MRP, shop control
4	Enterprise level	Corporate IT, ERP, sales, finance

- As we go up, the **time horizon grows** (milliseconds at device level ⟶ months at enterprise level).
- Lower levels need fast real-time control; higher levels deal with planning data.

10. Hardware Components for Automation and Process Control

Sensors

- Convert physical quantities into electrical signals: **temperature, pressure, flow, position, force, proximity**.
- Types: limit switch, photoelectric, inductive and capacitive proximity, encoders, thermocouples, strain gauges.

Actuators

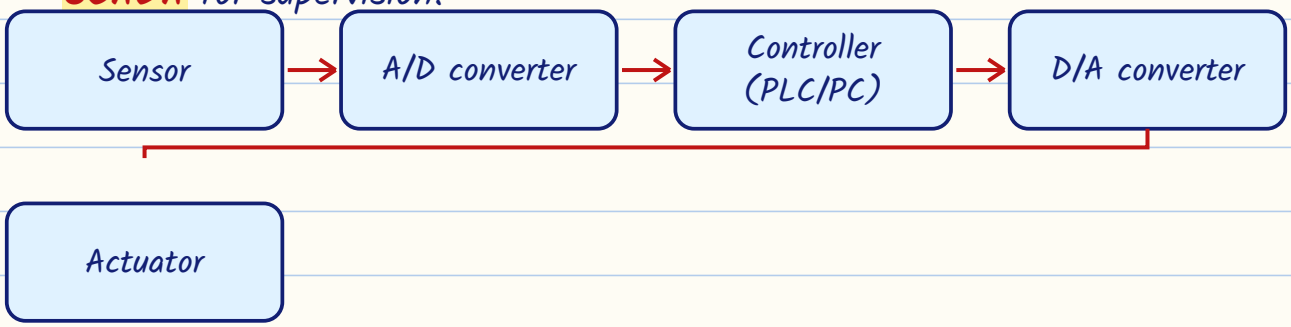
- Convert control signals into physical action: **solenoids, relays, motors (DC, AC, stepper, servo), pneumatic and hydraulic cylinders, valves**.

A/D and D/A Converters

- **ADC**: analogue signals from sensors ⟶ digital values for the controller.
- **DAC**: digital controller output ⟶ analogue voltage for the actuator.

Controllers

- **PLC** (Programmable Logic Controller): rugged, ladder-logic programming, widely used for sequence control.
- **PC based control**, microcontrollers, DCS (distributed control system), **SCADA** for supervision.



Process Control Types

- **Continuous control**: variables vary continuously (temperature, flow) -

PID control.

- **Discrete control**: on/off variables (limit switch, motor on) - PLC and relay logic.
- **Process industries** use continuous + batch; **discrete manufacturing** uses discrete control.

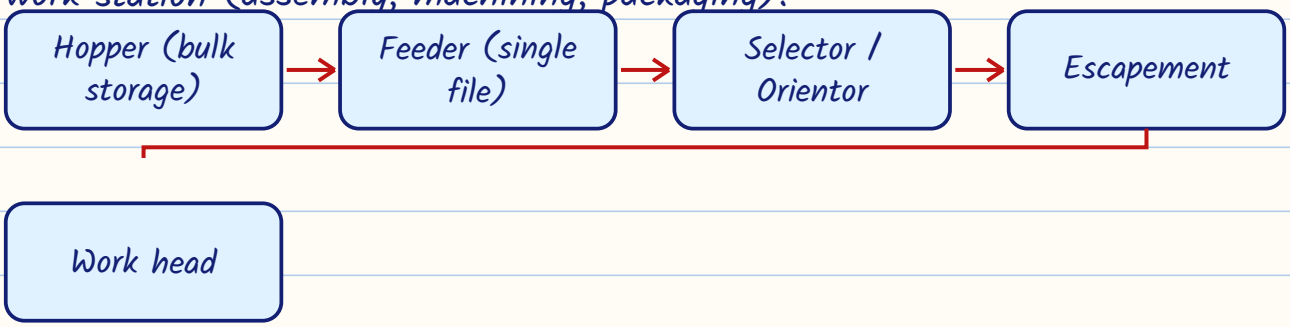
PLC Basics

- Parts: **CPU, input module, output module, power supply, programming device, memory**.
- Programming languages (IEC 61131-3): **ladder diagram**, function block diagram, structured text, instruction list.
- Scan cycle: read inputs ⟶ execute program ⟶ update outputs.

PLC replaces hard-wired relays; easy to modify, small, reliable, with self diagnostics.

11. Mechanical Feeders and Part Handling Devices

Automatic feeding supplies parts at the correct rate and **orientation** to the work station (assembly, machining, packaging).



Hoppers

- A **hopper** is a container that stores a bulk quantity of components in random orientation and feeds them to the feeder.
- Types: **fixed / gravity hoppers**, vibratory hoppers, rotary hoppers.
- Hopper should not jam or damage parts; capacity decides refilling interval.

Feeders

Feeders move parts from the hopper in the correct orientation.

Feeder	Working principle
Vibratory bowl feeder	Vibrating bowl with spiral track; parts climb and are oriented by selectors
Centreboard hopper feeder	Reciprocating centre blade lifts parts into a track
Rotary hopper feeder	Rotating drum / disc picks parts into chute
Magnetic feeder	Magnetic belt/drum for ferrous parts
Belt / roller feeder	Parts carried on conveyor belt
Reciprocating tube feeder	Tube reciprocates in the bulk and picks parts
External gate hopper	Reciprocating gate takes parts from a chute

Vibratory Bowl Feeder (most common)

- A circular bowl with a **helical track** inside; an electromagnet (or piezo) + leaf springs vibrate the bowl at about 50-120 Hz.
- Parts move **up the track with micro-hops**; unwanted orientations are rejected back into the bowl.
- Track tooling (**selectors**) gives the required orientation.
- Advantages: high rates, simple, no moving parts apart from the vibrator, quiet, handles many shapes.
- Limits: noise, not good for delicate or very sticky parts, tooling required for each part.

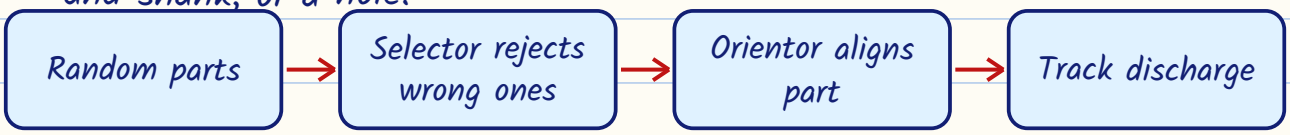
$$\text{Feed rate } f_r = n_p \times v / L$$

- Rate depends on track slope, vibration amplitude and frequency.

Selectors and Orientors

- **Selector**: removes unwanted parts / wrong orientation from the track.
  Examples: **wiper blade, pressure break, rails, notches**.

- **Orientor**: rotates or re-aligns a part into the correct position; works by **gravity and part geometry** using slots, pins, rotating guides.
- Typical orientors: gap (slot) orienting, rotating hook, drop through slot, tilted track.
- Method: use a feature such as an **asymmetric centre of gravity**, a head and shank, or a hole.



- **Example**: a screw / bolt: slot track lets the shank pass while the head rides on rails, so the bolt hangs head-up.
- Techniques: tilt track, wiper blade, pressure break, rail track, stripping gate.

Design rule: use the simplest geometry feature to orient; no complicated mechanism - reliability of a feeder is mostly in its tooling.

Escapements

- Escapement releases **one part at a time** to the work station at the right moment.
- Types: **reciprocating slide, rotary escapement, shuttle, gate type, star wheel**.

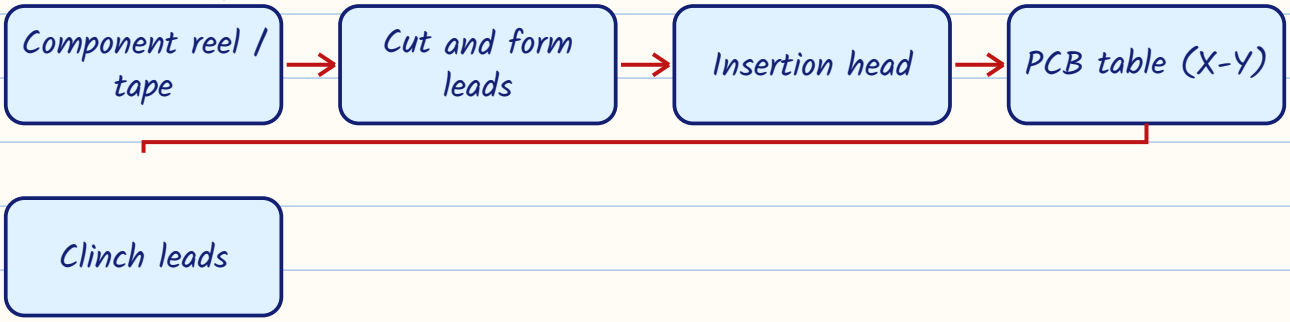
Magazines and Cartridges

- Parts are pre-oriented and stacked in a **magazine** (stack or tube). They need loading in advance but then give very reliable feeding for delicate parts.

Feeding system	Part orientation	Cost	Use
Bowl feeder	Automatic	Medium	Small parts, high rate
Magazine	Pre-oriented	Low	Delicate parts
Flexible feeder	Vision + robot	High	Many part types

12. High-Speed Automatic Insertion Devices

- Used in **electronics manufacturing** (PCB assembly) to insert / place components at high speed.
- Types of insertion:
  - **Radial lead insertion** - for components with two leads on the same side (capacitors, resistors).
  - **Axial lead insertion** - component leads on the same axis (resistors, diodes). The leads are cut and clinched.
  - **DIP insertion** - dual in-line IC packages.
  - **SMD / pick and place** - surface mount devices placed by a vacuum nozzle.



- A **CNC X-Y table** moves the PCB to the correct insertion position.
- Typical speeds: axial inserters 20,000+ components per hour; chip shooters up to 100,000+ cph.
- Features: **tape feeding, programmable sequence, vision alignment, error detection** (missing component).

Other Part Handling Devices

- **Pick-and-place units, parts feeders with robot, conveyors, walking beam, AGVs.**
- **Vacuum cups and magnetic grippers** to pick parts from the track.

Summary of feed system: Hopper (store) ⟶ Feeder (move) ⟶ Selector (reject) ⟶ Orientor (align) ⟶ Escapement (release) ⟶ Work head.

Quick Revision

- Automation: process without human help, power + program + control.
- Types: **fixed**, **programmable**, **flexible**.
- Production types: **job**, **batch**, **mass**.
- Levels of automation: device, machine, cell, plant, enterprise.
- Strategies: specialization, combined, simultaneous, integration, flexibility, handling, inspection, control, plant control, CIM.
- Feeder parts: hopper, feeder, selector, orientor, escapement.

Important Questions

- 1. Define automation. Explain types of automation with examples.
- 2. What are the reasons for automating? Discuss advantages and limitations.
- 3. Explain basic elements of an automated system with a neat sketch.
- 4. Explain types of production and compare them.
- 5. Discuss the ten strategies of automation.
- 6. Explain the levels of automation.
- 7. With a neat sketch explain the vibratory bowl feeder. What are selectors and orientors?
- 8. Write short notes on high-speed automatic insertion devices.